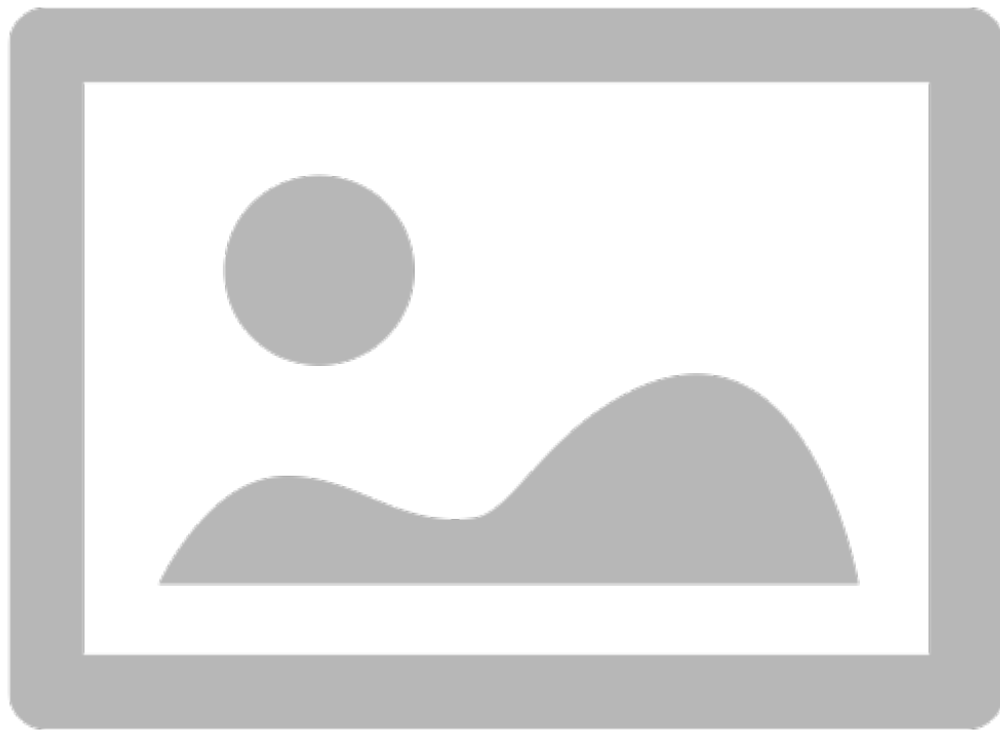


Project Proposal

Ryan Kembrey - 24464265



A capstone proposal on the development of a DFM workbench for FreeCAD
Engineering Research Preperation



Mechanical Engineering
University Of Technology Sydney
Australia

September 12, 2025

Contents

1	Introduction	5
2	Literature Review	6
2.1	Introduction	6
2.2	Manufacturing Domain	7
2.2.1	Core Principles of DFM for Injection Molding	7
2.2.2	Draft Angles	7
2.2.3	Wall Thickness	8
2.2.4	Undercuts	9
2.2.5	Corner Radii	9
2.2.6	Ribs	9
2.2.7	Bosses	10
2.2.8	Material	11
3	Synthesis and Problem Statement	12
3.0.1	Research Question	12
4	UI/UX Specification	13
4.1	Methodology	13
4.2	Case Studies of Competitor UX	14
4.2.1	DFMPro	14
4.2.2	SolidWorks DFMXpress	18
4.2.3	Autodesk Fusion 360	21
4.2.4	DFMA®	21
4.2.5	aPriori	21
4.3	Cross-Competitor Thematic Comparison	21
5	Proposed UX	22
6	Algorithmic Implementation	23
6.1	Draft Angle Algorithm	23
6.2	Undercut Algorithm	23
6.3	Thickness Algorithm	23
7	Code Infrastructure	24
7.1	Model-View-Controller (MVC)	24
7.2	A Plugin-Based Ecosystem (Strategy Pattern)	24
7.3	Data Flow (Pipes and Filters Architecture)	24
8	Methodology	25
	References	26
A	Project Plan	27
A.1	Roadmap	27
A.2	Resources	28
A.3	Team	28
A.4	Deliverables	28

[A.5 Risk Assessment](#) 28

List of Figures

2.1	Injection molding machine schematic. Credit: (E.I. du Pont de Nemours and Company, 2000)	6
2.2	Draft angle diagram	7
2.3	Uniform thickness at corners	8
2.4	Thickness transition guide	8
2.5	Boss design guide. Credit: (E.I. du Pont de Nemours and Company, 2000)	10
4.1	DFMPro on startup (DFMPro, 2022)	14
4.2	DFMPro analysis state widget (DFMPro, 2022)	14
4.3	DFMPro task panel tabs (DFMPro, 2022)	15
4.4	DFMPro highlighting a problematic hole (DFMPro, 2022)	15
4.5	DFMPro Rule Manager dialog (DFMPro, 2022)	16
4.6	DFMXpress when launched (Dassault Systèmes, 2024)	18
4.7	DFMXpress dialog box during analysis (Dassault Systèmes, 2024)	18
4.8	DFMXpress task panel in various states (Dassault Systèmes, 2024)	19
4.9	DFMXpress highlighting problematic geometry in the 3D view (Dassault Systèmes, 2024)	19

List of Tables

2.1	General guidelines for minimum draft angles in injection molding	7
2.2	General Draft Angle Guidelines for Common Materials (HLH Rapid, n.d.-a)	7
2.3	Recommended wall thickness for common injection molding materials (Xometry Team, n.d.) .	8
2.4	Summary of DFM Guidelines for Rib Design in Injection Moulding (HLH Rapid, n.d.-c) . . .	9
2.5	Summary of DFM Guidelines for Plastic Boss Design (HLH Rapid, n.d.-b)	10
2.6	Summary of Common Materials for Injection Moulding (HLH Rapid, n.d.-d)	11
3.1	Key DFM Checks and Justifications Identified from Literature Review	12
4.1	Categorisation of Competitor Software for UX Analysis	13
4.2	UI/UX grading system	13
4.3	Heuristic Evaluation for DFMPPro	17
4.4	Heuristic Evaluation for DFMPXpress	20
A.1	High-Level Project Roadmap and Deliverables	27
A.2	List of contributors	28
A.3	Table of deliverables	28

1 | Introduction

2 | Literature Review

2.1 Introduction

Design for manufacturing (DFM) is a crucial process in any products development. It is defined as the engineering practice of designing products for ease and economy of manufacturing. As computer-aided design (CAD) tools have become increasingly powerful, there has been a corresponding rise in automated tools to assist engineers in applying DFM principles to their designs. Whilst these solutions exist, there exists no free and open-source alternatives. The objective of the literature review is to conduct an analysis of the current landscape of DFM, in terms of existing automated tool and foundation standards that have long existed. Specifically, this review will focus on the manufacturing process of injection moulding, a well-established method of mass producing plastic parts with well defined standards and processes (Kauffer, 2011). The findings will be synthesised to justify and inform the development of an extensible DFM workbench for FreeCAD, an open-source, community-driven, parametric 3D CAD program.

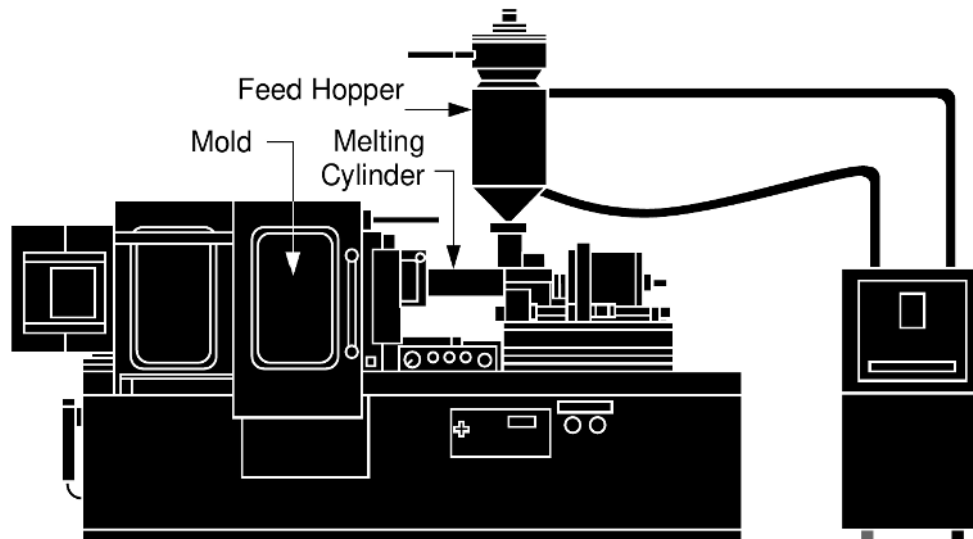


Figure 2.1: Injection molding machine schematic. Credit: (E.I. du Pont de Nemours and Company, 2000)

2.2 Manufacturing Domain

2.2.1 Core Principles of DFM for Injection Molding

When designing for injection molding, it is crucial to consider multiple factors that directly affect the manufacturability, quality, and cost of the final part. Adhering to established DFM principles mitigates production risks, reduces tooling complexity, and enhances the product's structural and cosmetic integrity.

2.2.2 Draft Angles

A draft angle is a slight taper applied to the faces of the part that are parallel to the mold's direction of pull (Oberg & Jones, 2020). This is a fundamental requirement to ensure that the part can be easily and cleanly ejected from the mold. Without an adequate draft, the friction between the part's surface and the mold wall is significant, which can lead to several manufacturing defects. There is a high risk of cosmetic damage, such as drag marks or scratches on the part's surfaces. Furthermore, the high forces required for ejection can introduce stress, leading to warping or even fracturing of the part (Oberg & Jones, 2020).

Figure 2.2 shows a diagram of an exaggerated draft angle. Notice the draft has been applied to both the inner and outer faces. While a general guideline is to apply a minimum draft of 1 to 2 degrees, the optimal angle depends on factors like wall depth, material choice, and surface texture. Textured surfaces, for instance, require a greater draft to prevent the finish from being scraped off during ejection (Oberg & Jones, 2020). The complexity of manually inspecting every surface on a complex geometric part makes this a time-consuming and error-prone task, highlighting the need for an automated analysis tool that can instantly identify non-compliant faces.

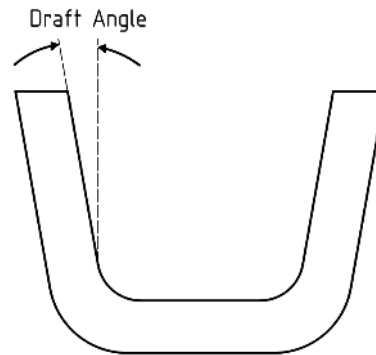


Figure 2.2: Draft angle diagram

Table 2.1: General guidelines for minimum draft angles in injection molding.

Application / Surface Description	Minimum Draft Angle
For “near-vertical” requirements	0.5°
General applications	2°
All shutoff surfaces	3°
Lightly textured faces	3°
Medium to heavily textured faces	5°+

Table 2.2: General Draft Angle Guidelines for Common Materials (HLH Rapid, n.d.-a)

Material	Minimum Draft (°)	Recommended Draft (°)
Nylon (PA)	0.0	1.0
Polyethylene (PE)	0.5	1.5
Polyvinyl Chloride (PVC)	0.5	1.5
Polypropylene (PP)	1.0	2.0
Polycarbonate (PC)	1.5	2.0

2.2.3 Wall Thickness

Maintaining a uniform wall thickness is one of the most critical principles in designing for injection molding (Oberg & Jones, 2020). Molten plastic is engineered to flow through the mold cavity and fill the part's geometry, but variations in thickness disrupt this process. Thicker sections cool much more slowly than thinner sections. This differential cooling causes uneven shrinkage (E.I. du Pont de Nemours and Company, 2000), which induces internal stresses in the part. These stresses can manifest as significant quality defects, including sink marks (small depressions on the surface opposite a thick feature), voids (internal bubbles), and warping.

The primary design goal is therefore to create a part with a consistent nominal wall thickness to promote uniform flow and cooling. Recommended wall thicknesses for various plastics can be found in Table 2.3. Large, solid features should be "cored out" and replaced with a network of thinner structural ribs to provide the necessary support without creating thick masses of material. If variations in thickness are absolutely unavoidable, the transition between sections should be gradual and smooth rather than abrupt.

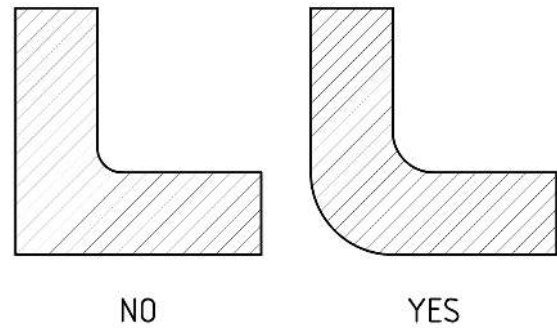


Figure 2.3: Uniform thickness at corners



Figure 2.4: Thickness transition guide

Table 2.3: Recommended wall thickness for common injection molding materials (Xometry Team, n.d.)

Material	Recommended wall thickness (mm)	Recommended wall thickness (in)
ABS	1.14 – 3.56 mm	.045 – .140 in
Acetal	0.76 – 3.05 mm	.030 – .120 in
Acrylic (PMMA)	0.64 – 12.70 mm	.025 – .500 in
Liquid Crystal Polymer	0.76 – 3.05 mm	.030 – .120 in
Long-Fiber Reinforced Plastics	1.91 – 27.94 mm	.075 – 1.100 in
Nylon	0.76 – 2.92 mm	.030 – .115 in
PC (Polycarbonate)	1.02 – 3.81 mm	.040 – .150 in
Polyester	0.64 – 3.18 mm	.025 – .125 in
Polyethylene (PE)	0.76 – 5.08 mm	.030 – .200 in
Polyphenylene Sulfide (PSU)	0.51 – 4.57 mm	.020 – .180 in
Polypropylene (PP)	0.89 – 3.81 mm	.035 – .150 in
Polystyrene (PS)	0.89 – 3.81 mm	.035 – .150 in
Polyurethane	2.03 – 19.05 mm	.080 – .750 in

Visualizing wall thickness across an entire part is non-trivial, and identifying areas of unwanted material concentration or thinness is a key challenge that software analysis is uniquely equipped to address.

2.2.4 Undercuts

An undercut is a feature in a part design, such as a side hole or a snap-fit latch, that would prevent its direct ejection from a simple two-part mold. Because these features are not aligned with the pull direction, they would lock the part within the mold cavity.

Manufacturing parts with undercuts requires significantly more complex and expensive tooling. These molds must incorporate moving components known as side-actions or lifters. These mechanisms slide into place to form the undercut feature and must then retract before the main mold opens for part ejection. The addition of these mechanisms dramatically increases the initial cost and lead time for mold fabrication, adds complexity to its ongoing maintenance, and can increase the production cycle time for each part. As a result, a key DFM objective is to avoid or eliminate undercuts wherever possible to reduce overall manufacturing cost and complexity. An automated DFM tool would provide much insight to an engineer during the design phase, such that the undercuts could be eliminated before being sent off for manufacturing review by the supplier.

2.2.5 Corner Radii

Sharp internal corners in a part's design are a source of multiple issues in injection molding. From a flow perspective, it is difficult for molten plastic to navigate and properly fill sharp, 90-degree corners. More critically, sharp internal corners create a high stress concentration. This concentration of stress acts as a weak point, significantly compromising the part's mechanical strength and making it susceptible to failure when placed under load.

To mitigate these risks, all corners should be designed with a generous radius. Applying fillets to all internal and external corners distributes stress more evenly, greatly improving the part's durability. A generous radius also improves the plastic flow within the mold. Furthermore, rounded corners are easier and less costly to machine into the mold cavity, as rotating toolbits naturally create filleted features.

2.2.6 Ribs

Ribs are a reinforcing feature that improve a part's structural rigidity. They should only be used when crucial for the parts performance or to improve mold flow to another feature. Excessive use of ribbing in a part can cause warping of the plastic and points of high stress concentration.

Table 2.4: Summary of DFM Guidelines for Rib Design in Injection Moulding (HLH Rapid, n.d.-c)

Design Parameter	Recommended Value	Justification
Rib Height	$\leq 3x$ Nominal Wall Thickness	Prevents incomplete filling (short shots) and breakage during part ejection.
Rib Thickness	40% – 60% of Nominal Wall Thickness	Critical for preventing cosmetic sink marks on the opposite surface due to uneven cooling.
Base Radius	0.5x – 1x Nominal Wall Thickness	Reduces high stress concentration at the base and improves molten plastic flow.
Rib Spacing	$\geq 2x$ Nominal Wall Thickness	Ensures the mould steel between ribs is strong enough and promotes uniform cooling.
Draft Angle	$\geq 0.5^\circ$ per side	Facilitates a clean release of the part from the mould without drag marks.

2.2.7 Bosses

Bosses are often used as a mounting feature for external parts or as reinforcement around a hole.

Table 2.5: Summary of DFM Guidelines for Plastic Boss Design (HLH Rapid, n.d.-b)

Design Parameter	Recommended Value	Justification
Boss Placement	Distribute placement. Close to thicker walls. On interior of part.	Avoids concentrating stress and heat in one area, reducing risk of warp and defects.
Wall Thickness	$\leq 60\%$ of Nominal Wall	Prevents cosmetic sink marks and internal voids caused by excess material.
Support Features	Use gussets, ribs, or connect to a main wall	For structural support.
Height-to-Diameter Ratio	Height $< 3\times$ Outer Diameter	Limits mechanical stress on the boss and ensures it can be filled properly during moulding.
Thread Engagement	Length $\geq 60\%$ of Screw Thread	Ensures sufficient surface area for the fastener to grip securely.
Boss Spacing	$\geq 2\times$ Nominal Wall Thickness	Maintains the integrity of the mould steel between features and promotes uniform cooling.
Draft (Inner Diameter)	$\geq 0.25^\circ$	Allows the core pin to be removed easily without damaging the inner surface.
Draft (Outer Diameter)	$\geq 0.5^\circ$	Facilitates a clean release of the part from the mould cavity without drag marks.
Base Radius	$0.25\times - 0.5\times$ Nominal Wall	Critically reduces stress concentration at the base, preventing the boss from breaking under load.

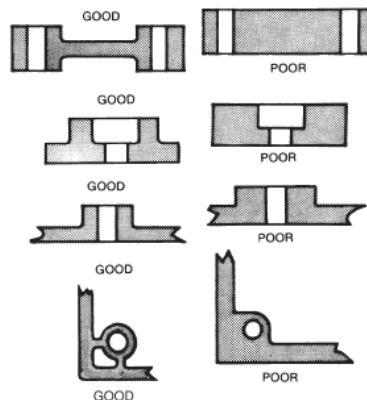


Figure 2.5: Boss design guide. Credit: (E.I. du Pont de Nemours and Company, 2000)

2.2.8 Material

Table 2.6: Summary of Common Materials for Injection Moulding (HLH Rapid, n.d.-d)

Material (Acronym)	Key Properties	Common Applications
PMMA (Acrylic)	Transparency, High Gloss, UV Resistance	Lenses, Lighting, Transparent Parts
PC (Polycarbonate)	Impact Resistance, High Strength, Clarity	Safety Equipment, Machinery Guards
ABS	General Purpose, Low Cost, Good Aesthetics	Consumer Products, Housings, Automotive
PP (Polypropylene)	Low Cost, Flexibility, Chemical Resistance	Living Hinges, Containers, Power Tools
PA (Nylon)	High Strength, Durability, High Temperature	Gears, Screws, Mechanical Parts, Snap-fits
PEEK	High Performance, High Cost, Extreme Temperature	Aerospace, Bio-medical, Electrical
POM (Acetal)	High Strength, Low Friction, Wear Resistance	Gears, Bearings, Bushings, Moving Parts
TPE	Rubber-like, Flexibility, Overmoulding	Soft-touch Grips, Seals, Flexible Parts
TPU	Hard Rubber, Elasticity, Abrasion Resistance	Gaskets, Wearables, Medical Tubing
<i>Common Material Blends</i>		
PC/ABS	High Strength, Heat Resistance, Processability	Automotive, Electronic Housings
PC/PBT	High Strength-to-Weight, Toughness, Chemical Resistance	Industrial Equipment, Electronics Covers

3 | Synthesis and Problem Statement

The literature clearly states a need for automation of checking designs against DFM principles. Specifically, injection moulding has presented itself as the perfect candidate for such automation against existing standards. Table 3.1 lists each DFM principle that should be checked in an automated DFM tool for injection moulding.

Table 3.1: Key DFM Checks and Justifications Identified from Literature Review

Automated Check	Justification / Core Purpose
Draft Angle	Ensures clean part ejection from the mould, preventing cosmetic defects (drag marks) and mechanical stress.
Wall Thickness	Promotes uniform cooling and plastic flow to prevent sink marks, voids, and warping.
Undercuts	Identifies features that would require complex and costly side-action tooling, allowing for cost-saving redesigns.
Corner Radii	Mitigates stress concentration at internal corners to improve the part's mechanical strength and durability.
Rib Design	Validates that thin, structural ribs are used correctly to add rigidity without causing cosmetic defects like sink marks.
Boss Design	Validates that mounting and alignment bosses are designed to be strong while preventing sink marks and internal voids.

3.0.1 Research Question

A research question was formulated to guide the research and development of an automated DFM workbench for FreeCAD.

Research Q1:

“Can a modular and extensible design for manufacturing workbench be created for FreeCAD, such that a design can be run against a set of defined rules and generate iterative useful feedback.”

Research Q2:

“Can the functional gap between proprietary CAD platforms and the open-source FreeCAD ecosystem be addressed by developing a modular and extensible design for manufacturing workbench, such that a design can be run against a set of pre-defined or user-defined rules and generate iterative useful feedback.”

Research Q3:

*“To what extent can a DFM workbench for FreeCAD be implemented to function not merely as a validation tool, but as an educational platform that actively improves a designer's understanding of manufacturing principles by integrating contextual help documentation and a guided, interactive feedback loop that explains the **why** behind each detected issue?”*

4 | UI/UX Specification

4.1 Methodology

To establish a robust foundation for the proposed user experience of the FreeCAD DFM Workbench, a competitive analysis of the existing DFM software landscape was conducted. This analysis serves to identify established UX patterns, evaluate their effectiveness, and synthesise key principles that can inform the workbench's design. The methodology involved identifying a representative selection of market competitors and performing a heuristic evaluation of their user workflows based on publicly available documentation, tutorials, and product demonstrations. Five competitors were chosen for evaluation, listed in Table 4.1.

Table 4.1: Categorisation of Competitor Software for UX Analysis

Platform / Software	Developer	Platform Type
DFMPro	HCL Technologies	Integrated Add-in
SOLIDWORKS DFMXpress	Dassault Systèmes	Integrated Add-in
Autodesk Fusion 360	Autodesk, Inc.	Holistic Environment
DFMA®	Boothroyd Dewhurst, Inc.	Standalone Platform
aPriori	aPriori Technologies	Standalone Platform

To inform the design of the FreeCAD DFM Workbench, a formal heuristic evaluation was conducted to systematically analyse the UI/UX of key competitor software. Each platform was rigorously assessed against a comprehensive checklist of criteria covering the entire user workflow, from initial launch to results analysis and customisation. To ensure a consistent and objective evaluation, a four-point rating scale (see Table 4.2) was employed, with every rating substantiated by a concise written justification. This process yielded a set of standardised evaluation tables that serve as the primary data for the subsequent cross-competitor comparison.

Table 4.2: UI/UX grading system

Grade	Meaning
Pass +	The feature is implemented exceptionally well.
Pass -	The feature is present and works effectively.
Fail +	The feature exists but is flawed, cumbersome, or incomplete.
Fail -	The feature is completely missing.

A case study was carried out for each competitor. Each analysis consists of a user-journey, expressed through screenshots of the UI and explanations of interactive process. A heuristic evaluation is then conducted for each competitor in the form of a table. This aims to turn the subjective study into objective guidelines to aid the development of the FreeCAD DFM Workbench.

4.2 Case Studies of Competitor UX

4.2.1 DFMPPro

DFMPPro (HCL Technologies, 2024) is an integrated add-in available for SOLIDWORKS, CATIA V5, NX, Creo Parametric and 3DEXperience. This UX analysis will be of DFMPPro within Creo Parametric.

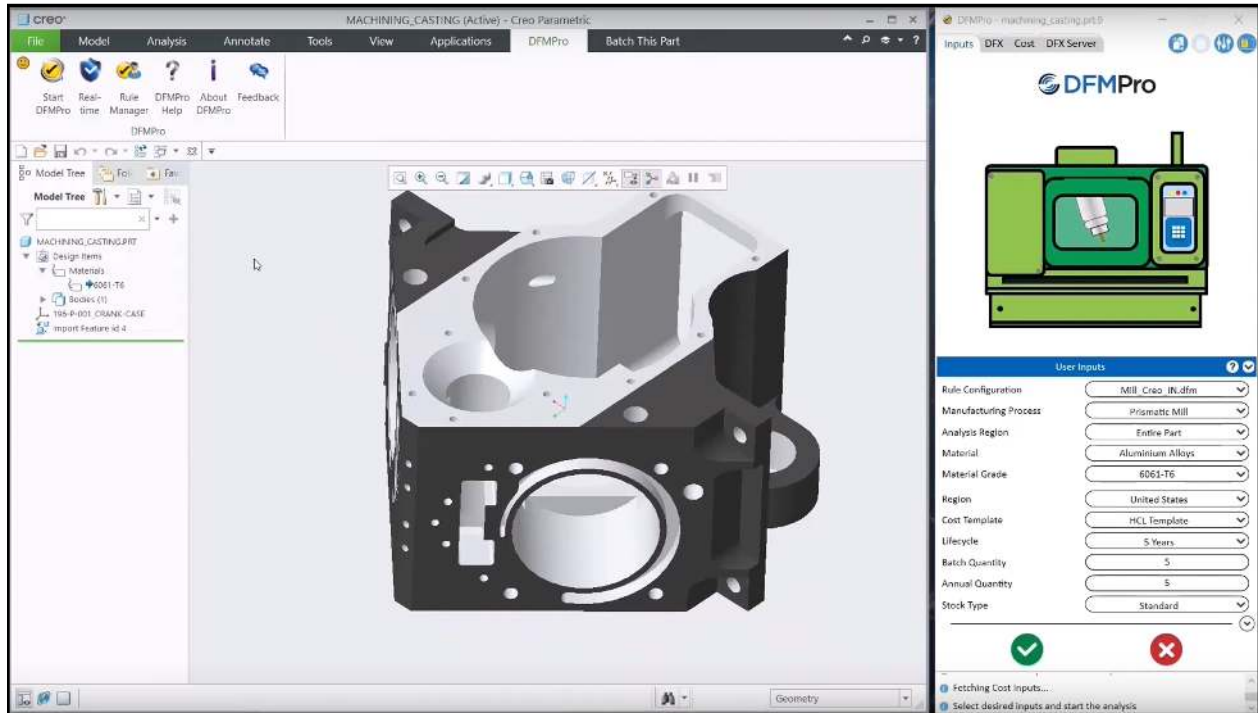


Figure 4.1: DFMPPro on startup (DFMPPro, 2022)

The DFMPPro tool is accessed from the **DFMPPro** tab inside of Creo, which opens the DFMPPro toolbar when clicked. Here six tools are presented to the user. The main panel is launched from accessing the **Start DFMPPro** tool. The start task panel is separate from the main Creo window, and presents four tabs at the top of the window; **Inputs**, **DFX**, **Cost** and **DFXServer**. The first three of which are shown in Figure 4.3.

In the **Inputs** panel (Figure 4.3a), the user is presented with a list of drop-downs menus to configure before running the analysis. DFMPPro de-couples the management of rule configuration into user-defined files, editable in the Rule Manager dialog (Figure 4.5). The **Inputs** panel allows for configuration of the manufacturing process, analysis region, material, material grade, region, cost template, lifecycle, batch quantity, annual quantity and stock type. Once the analysis is setup, the green tick icon can be clicked to run the study. An analysis state widget now appears in the now empty task panel (4.2).

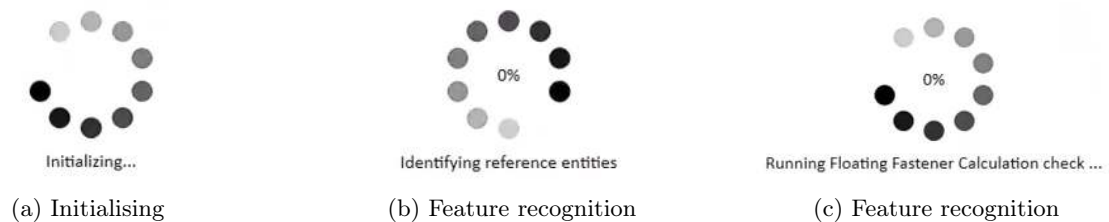


Figure 4.2: DFMPPro analysis state widget (DFMPPro, 2022)

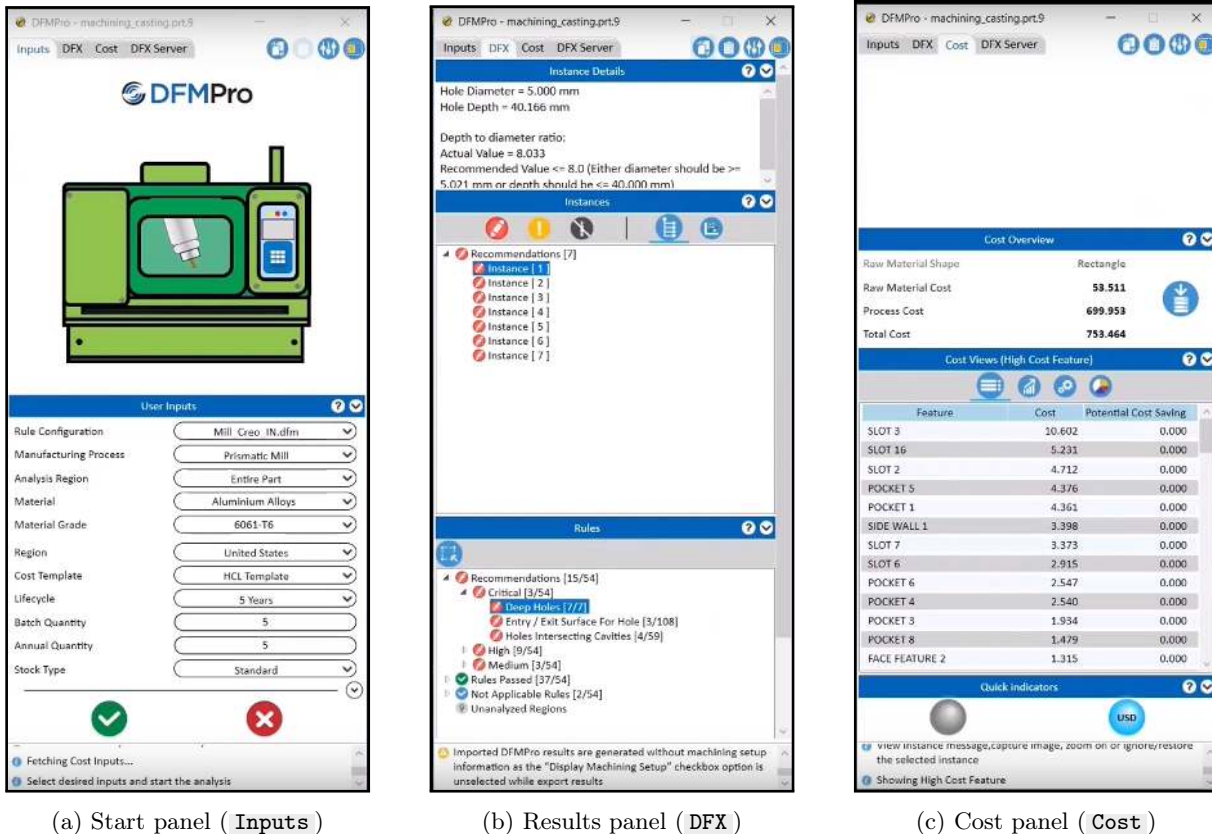


Figure 4.3: DFMPPro task panel tabs (DFMPPro, 2022)

Once the analysis is complete, DFMPPro automatically switches to the results tab (DFX Figure 4.3b). The panel consists of three main sections; the Instance Details, Instances and Rules. For any given analysis, there may be a large number of feedback points (instances). DFMPPro de-couples the instances from their respective rule by separating them into different groupings within the panel. This way, the user selects the Rule they are wishing to view feedback for in the Rules group, and the Instances window is populated with a list of Recommendations. When a recommendation is clicked, the 3D view automatically zooms to the problematic geometry.

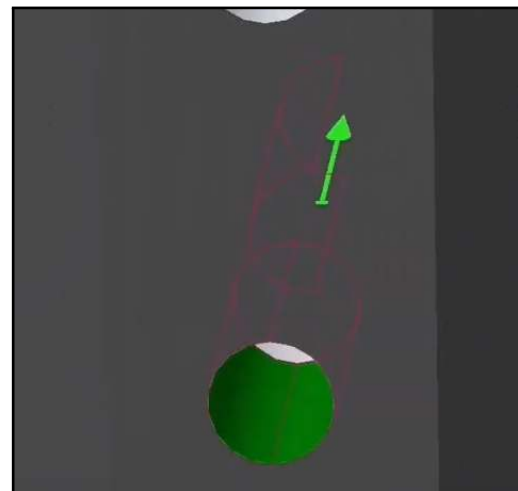


Figure 4.4: DFMPPro highlighting a problematic hole (DFMPPro, 2022)

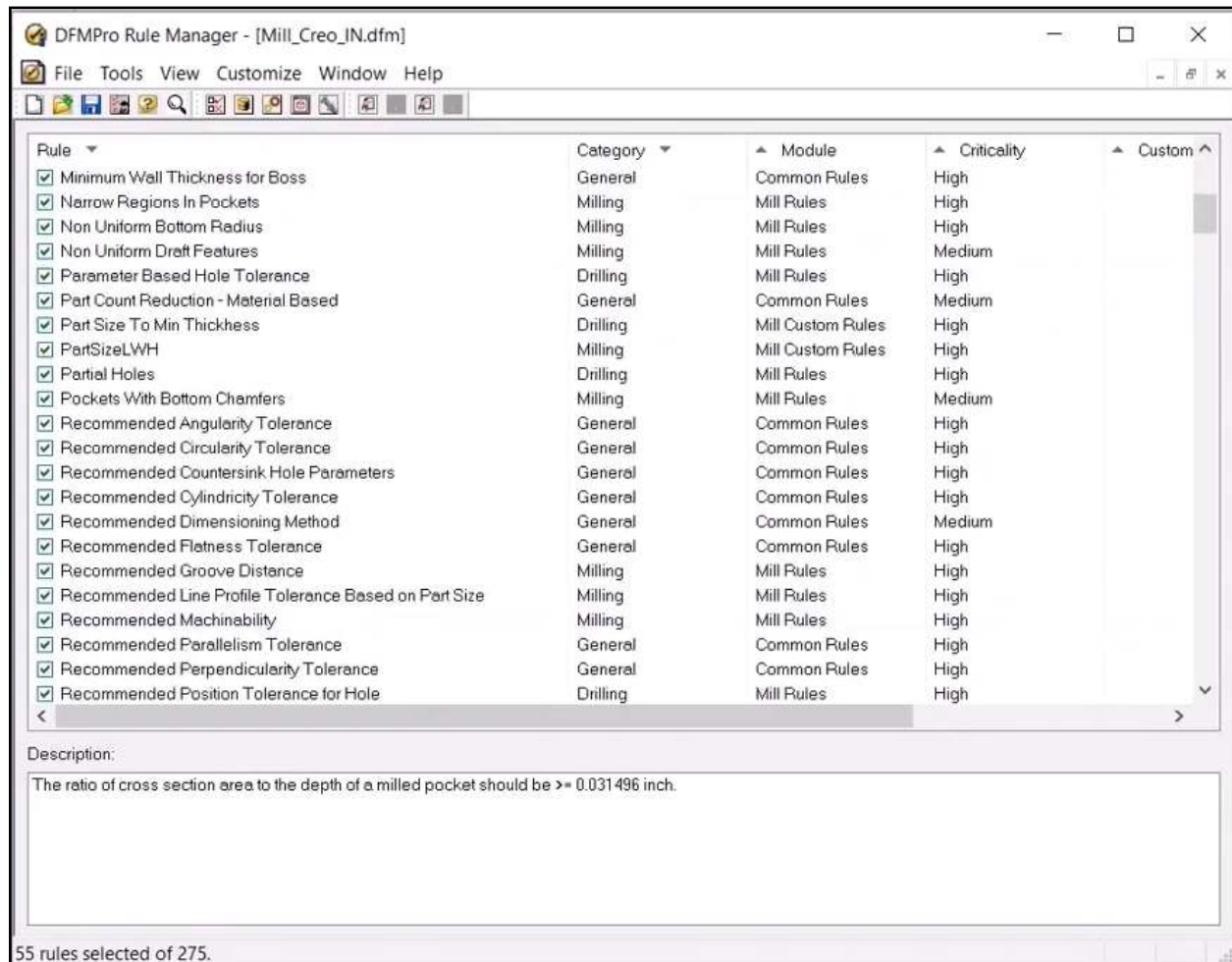


Figure 4.5: DFMPPro Rule Manager dialog (DFMPPro, 2022)

Table 4.3: Heuristic Evaluation for DFMPPro

Criterion	Rating	Justification
1. Initiation & Access		
Entry Point Discovery		
Entry Point Clarity		
Host UI Consistency		
2. Configuration & Setup		
Primary Interface Paradigm		
Guided Workflow		
Geometric Input Method		
Granularity of Check Selection		
Persistence of Settings		
3. System Feedback		
Execution Status Feedback		
UI Responsiveness During Analysis		
Cancellability of Operation		
4. Results Presentation		
Results Hierarchy		
Categorisation of Findings		
Readability of Results		
Actionability of Information		
Use of Colour and Icons		
5. Interaction with Results		
Automated Camera Navigation		
Geometry Highlighting Clarity		
Interaction Responsiveness		
Use of Tooltips/Pop-ups		
6. Customization & Extensibility		
Rule Management Interface		
Accessibility of Customization		
Extensibility (New Rules/Materials)		
7. Overall Heuristics		
Aesthetic		
Error Prevention		
Help and Documentation Access		

4.2.2 SolidWorks DFMXpress

SolidWorks DFMXpress (Dassault Systèmes, 2024) is an integrated DFM analysis add-in that runs inside of SolidWorks (Morrogh, 2023). Developed by HCL Technologies, the company behind the professional DFM-Pro suite, DFMXpress serves as a more streamlined and introductory version of the tool.

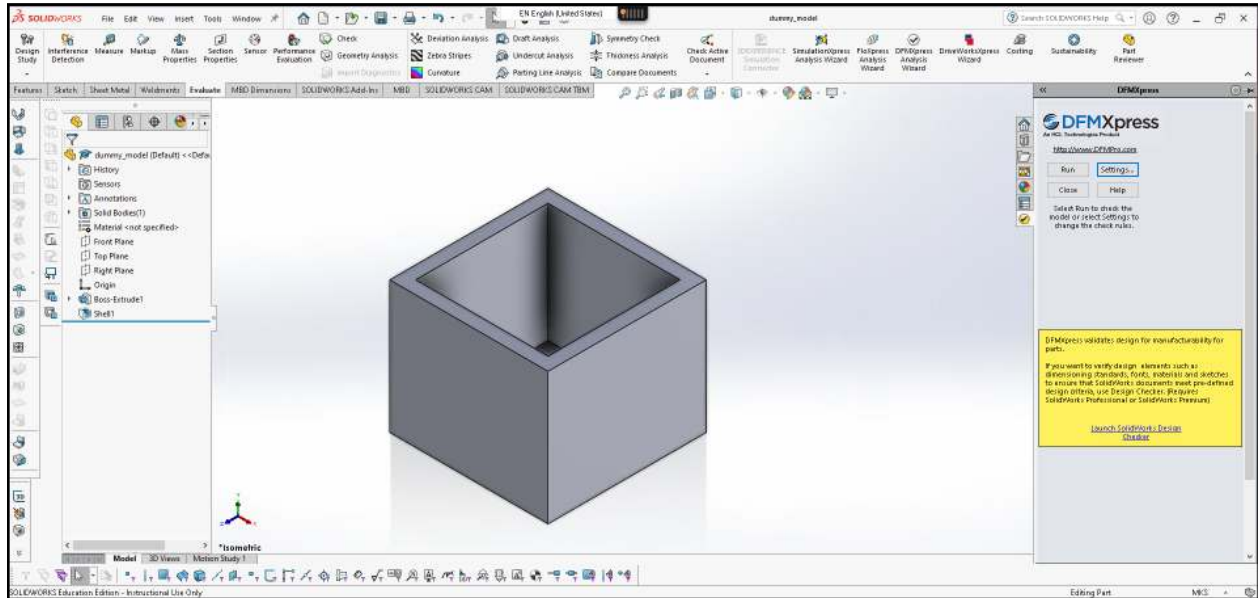


Figure 4.6: DFMXpress when launched (Dassault Systèmes, 2024)

The DFMXpress tool is initiated from the **Evaluate** tab in the SOLIDWORKS command ribbon. This action launches a dedicated task panel on the right side of the interface, which presents four primary commands: **Run**, **Settings**, **Close**, and **Help** (Figure 4.8a). Clicking the **Settings** button allows the user to configure the analysis parameters for a specific manufacturing process (Figure 4.8b). Once configured, the **Run** button executes the analysis on the active SOLIDWORKS part, while the **Help** button provides access to the user documentation in a web browser.

During analysis, a small dialog box appears, showing the progression of the computations (Figure 4.7). Once the analysis has finished, the task panel is then populated with a list of feedback instances (Figure 4.8c).

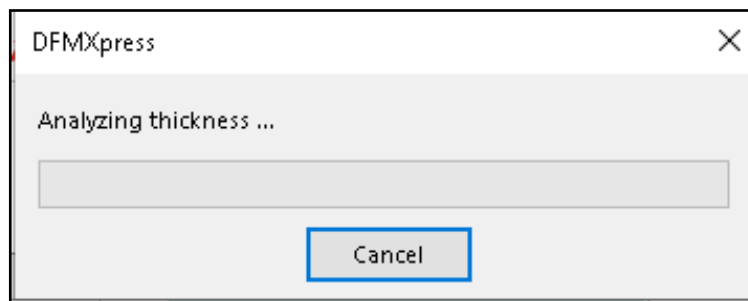
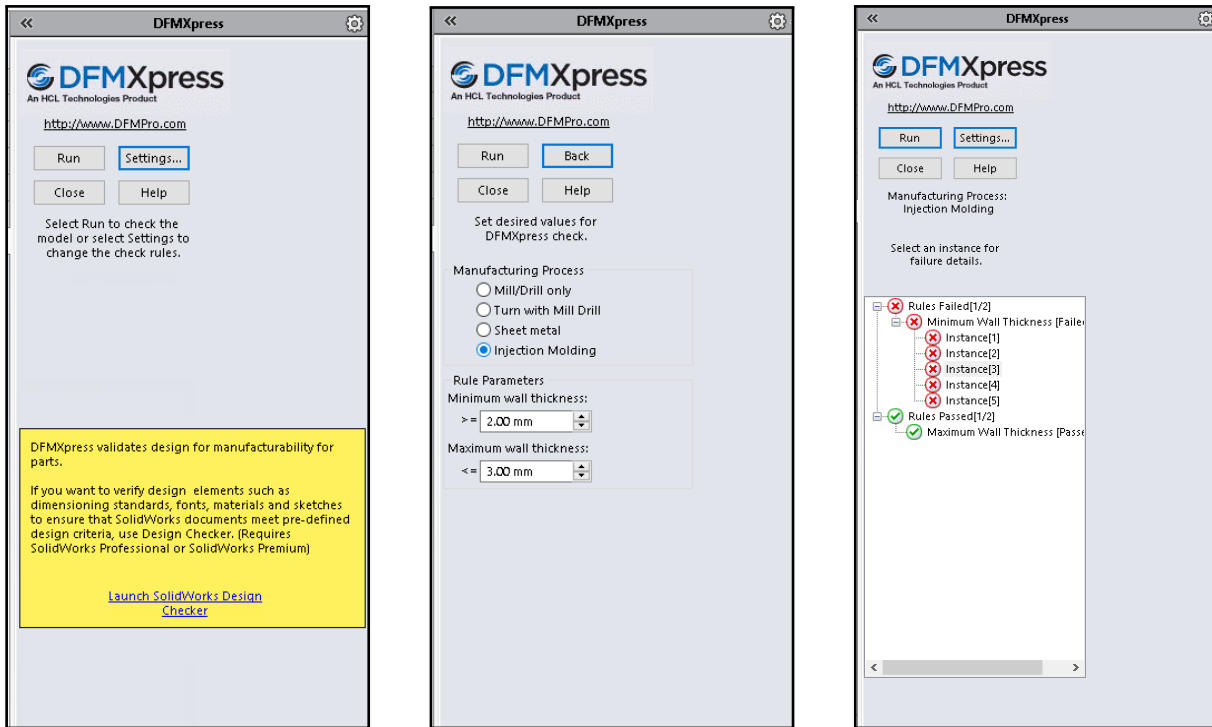


Figure 4.7: DFMXpress dialog box during analysis (Dassault Systèmes, 2024)

Selecting a specific finding in the results panel triggers an immediate, dual-feedback response: the system highlights the corresponding geometry in the 3D view, while a tooltip appears with a concise explanation of the failure (Figure 4.9).



(a) When started

(b) Injection molding settings

(c) Results

Figure 4.8: DFMXpress task panel in various states (Dassault Systèmes, 2024)

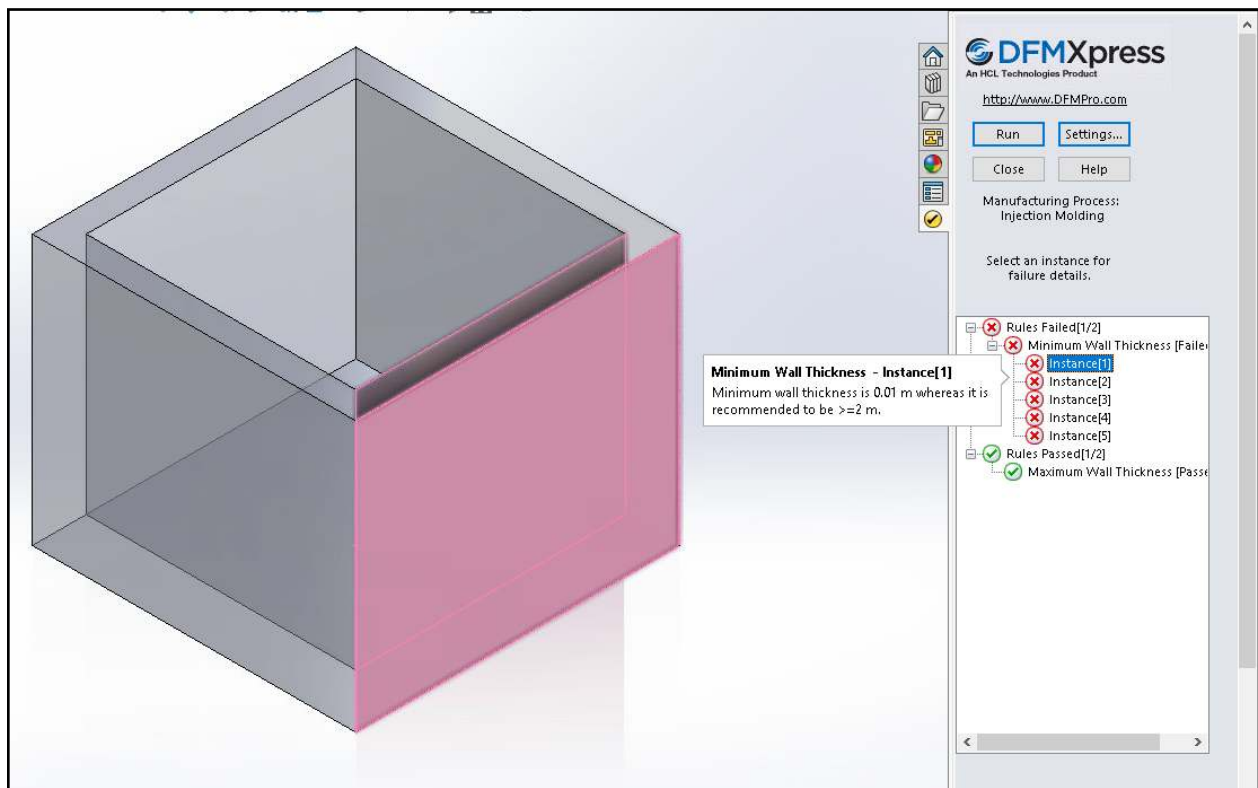


Figure 4.9: DFMXpress highlighting problematic geometry in the 3D view (Dassault Systèmes, 2024)

Table 4.4: Heuristic Evaluation for DFMXpress

Criterion	Rating	Justification
1. Initiation & Access		
Entry Point Discovery	Pass -	In right tab, lost amongst other tools
Entry Point Clarity	Pass -	In Evaluate tab in SOLIDWORKS
Host UI Consistency	Fail -	Different colours, poor element placements
2. Configuration & Setup		
Primary Interface Paradigm	Pass -	Primary idea good, execution bad
Guided Workflow	Fail +	Minimal guidance in task panel
Geometric Input Method	Fail -	Not implemented
Granularity of Check Selection	Fail -	Not implemented
Persistence of Settings	Fail +	Persistence on panel close, but not program close
3. System Feedback		
Execution Status Feedback	Pass -	Pop-up status dialog, but could use more info
UI Responsiveness During Analysis	Pass +	Responds
Cancellability of Operation	Pass +	Cancellable in pop-up status dialog
4. Results Presentation		
Results Hierarchy	Fail -	No hierarchy system
Categorisation of Findings	Pass +	Categorised in list tree
Readability of Results	Pass -	Task panel UI is static and needs horizontal scrolling
Actionability of Information	Fail +	No tips on how to fix in CAD
Use of Colour and Icons	Pass -	Colour in 3D view good. Panel bland
5. Interaction with Results		
Automated Camera Navigation	Fail -	Not implemented
Geometry Highlighting Clarity	Pass +	Very clear with use of colours in 3D view
Interaction Responsiveness	Pass -	Panel changes depending on selection
Use of Tooltips/Pop-ups	Pass -	Some tooltips
6. Customization & Extensibility		
Rule Management Interface	Fail +	Exists in task panel, but limited
Accessibility of Customization	Fail +	Accessible directly in task panel
Extensibility (New Rules/Materials)	Fail -	No implemented
7. Overall Heuristics		
Aesthetic	Fail -	Outdated UI. Poor widget placements
Error Prevention	Pass -	Some error prevention when defining rules
Help and Documentation Access	Pass -	Help accessible through a link to website

4.2.3 Autodesk Fusion 360

4.2.4 DFMA®

4.2.5 aPriori

4.3 Cross-Competitor Thematic Comparison

5 | Proposed UX

6 | Algorithmic Implementation

6.1 Draft Angle Algorithm

6.2 Undercut Algorithm

6.3 Thickness Algorithm

7 | Code Infrastructure

7.1 Model-View-Controller (MVC)

7.2 A Plugin-Based Ecosystem (Strategy Pattern)

7.3 Data Flow (Pipes and Filters Architecture)

8 | Methodology

References

- Dassault Systèmes. (2024). *SOLIDWORKS DFMXpress*. Author. Retrieved 2025-09-07, from <https://www.solidworks.com/product/solidworks-3d-cad> (Integrated software feature in SOLIDWORKS 2025)
- DFMPro. (2022, dec 13). *Digital transformation in manufacturing through design for manufacturing*. Retrieved 2025-09-07, from <https://youtu.be/adK5r0BYI6I>
- E.I. du Pont de Nemours and Company. (2000). *General design principles for dupont engineering polymers* (Tech. Rep. No. H-76838). Wilmington, DE: DuPont Engineering Polymers.
- HCL Technologies. (2024). *Dfmpro*. HCL Technologies. Retrieved 2025-09-07, from <https://dfmpro.com>
- HLH Rapid. (n.d.-a). *Draft Angle Design Guide & Best Practices: How To Design For Better Mould Release*. Retrieved 2025-09-10, from <https://hlhrapid.com/knowledge/draft-angles-improving-part-mouldability/>
- HLH Rapid. (n.d.-b). *How to Design Bosses for Plastic Parts: Injection Moulding Design Guide*. Retrieved 2025-09-10, from <https://hlhrapid.com/knowledge/bosses-design-guide-injection-moulding/>
- HLH Rapid. (n.d.-c). *How to Design Ribs for Plastic Parts: Injection Moulding Design Guide*. Retrieved 2025-09-10, from <https://hlhrapid.com/knowledge/rib-design-guide-injection-moulding/>
- HLH Rapid. (n.d.-d). *Injection Moulding Material Selection Guide - Choosing a Plastic For Your Project*. Retrieved 2025-09-10, from <https://hlhrapid.com/blog/choosing-a-material-for-plastic-injection-moulding/>
- Kauffer, P. H. (2011). *Injection molding : process, design, and applications* (1st ed. ed.). New York: Nova Science Publishers.
- Morrogh, R. (2023, 12 18). *Make it Manufacturable – Design for Manufacturing in SOLIDWORKS with DFMXpress*. Retrieved 2025-09-07, from <https://www.javelin-tech.com/blog/2023/12/make-it-manufacturable-design-for-manufacturing-in-solidworks-with-dfmexpress/>
- Oberg, E., & Jones, F. D. (2020). *Machinery's handbook* (31st ed.). New York: Industrial Press.
- Xometry Team. (n.d.). *Design guidelines and best practices for injection molding* (Tech. Rep.). Xometry.

A | Project Plan

A.1 Roadmap

Table A.1: High-Level Project Roadmap and Deliverables

Phase	Name	Timeline	Key Deliverables
1	Proof-of-Concept & Research	Aug - Sep 2025	• Stakeholder needs analysis. • Artefact <i>System Requirements Specification</i> • Artefact <i>Market Competitor Specification</i> • Basic draft check. Prints to terminal.
2	Prototype	Oct 2025	• Artefact <i>Algorithmic Implementation Specification</i> • Test suite of .step models for objective testing of checks • Fully functioning draft and undercut checks • Can be run from setup task panel GUI. Results in terminal. • Artefact <i>Code Architecture Specification</i>
3	Minimum Viable Product (MVP)	Nov 2025	• Results task panel functioning • Highlight problem geometry in 3D view • Developer and user documentation
4	V0.1 Release	Dec 2025 - Jan 2026	• "Code Architecture Specification" verdict implemented fully • Thickness check • Local HTML help pages accessible from results panel • Auto-zooming to problem geometry in 3D view
5	Stretch Goals	Feb - June 2026	• Boss, fillet, rib checks • Secondary pull directions • Custom DFM standards, editable in GUI, saved locally. • Export report (spreadsheet in .odt format)

A.2 Resources

A.3 Team

Refer to Table [A.2](#) for a list of all contributing team members. A contribution can be defined as any of the following; feedback, design discussions, review, code creation, icon creation, testing.

FreeCAD has multiple working groups, each with their own specialisation. These are:

- Design Working Group (DWG)
- CAD Working Group (CWG)
- Code Quality Working Group (CQWG)

Table A.2: List of contributors

Contributer	Role
Ryan Kembrey	Capstone student
Anna Lidfors-Lindqvist	Supervisor
Benjamin Nauck	Co-Supervisor, DWG, CQWG
@Pierreporte	CWG, DWG
@Max	CWG, DWG
@Obelisk	DWG

A.4 Deliverables

Table A.3: Table of deliverables

Deliverable	Due Date	Description
Literature Review	Sep 2025	PDF report summarising the existing research and identification of a research question
Project Proposal	Oct 2025	PDF report outlining the proposed capstone project.
MVP	TBD	
FreeCAD DFM Workbench	TBD	Hosted on GitHub, installable via the Addon Manager
Code documentation	TBD	
User documentation	TBD	

A.5 Risk Assessment